

# Demonstrating Diff-MSTC: A Controllable and Context-Aware AI System for Multitrack Mixing in Digital Audio Workstation Cubase

Soumya Sai Vanka  
Centre for Digital Music  
Queen Mary University of London  
London, United Kingdom  
s.s.vanka@qmul.ac.uk

Jean-Baptiste Rolland  
Steinberg Media Technologies GmbH  
Hamburg, Germany  
jb.rolland@steinberg.de

György Fazekas  
School of Electronic Engineering and  
Computer Science  
Queen Mary University of London  
London, United Kingdom  
g.fazekas@qmul.ac.uk

## CCS Concepts

• **Applied computing** → **Sound and music computing**; • **Computing methodologies** → *Neural networks*; • **Human-centered computing** → **User interface design**; *User centered design*.

## Keywords

Intelligent Music Production, Music Mixing, HAI, Creativity, Audio Effects, User-Centric Design

### ACM Reference Format:

Soumya Sai Vanka, Jean-Baptiste Rolland, and György Fazekas. 2025. Demonstrating Diff-MSTC: A Controllable and Context-Aware AI System for Multitrack Mixing in Digital Audio Workstation Cubase. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3706599.3721167>

## 1 Abstract

Intelligent music production has garnered attention for over a decade. Despite this interest, there are few prototypes functional in practical workflows, especially those that consider context, essential to the mixing process. In this demonstration, we introduce **Diff-MSTC**, a deep learning-based multitrack mixing system, developed with a user-centric approach, that is both controllable and context-sensitive, and seamlessly integrates within a popular digital audio workstation. This system adapts to variations in track number and order and produces a mix in the style of provided reference song. We explore its application, provide a subjective perspective from both a researcher and user standpoint, and delve into the questions it raises regarding usability, effectiveness, workflow transformation, and the need for further investigation into the human-centered design of these tools.

## 2 Introduction and Background

Music mixing is a critical task in music post-production, shaping the final sonic quality and character of a piece by balancing gain levels, addressing frequency masking, and crafting the dynamic and

tonal identity using audio effect plugins or traditional analog equipment [7, 12]. As both a technical and creative process, it demands expertise in sound, audio engineering, and signal processing. The emerging field of Intelligent Music Production (IMP) [6] explores assistive and automated systems for music production. Over the past 15 years, the subfield of automatic mixing has evolved from rule-based systems to machine learning approaches [5, 13, 21], with deep learning now showing great potential for advancing multitrack mixing.

Deep learning approaches to mixing can be broadly categorized into two regimes: parameter estimation (PE) and direct transformation (DT) [20]. As shown in Figure 2, the DT approach treats mixing as a black-box process, taking multitracks as input and predicting the final mix in the audio or frequency domain. While this method can yield results, it often lacks interpretability and user control [10, 11]. In contrast, the PE approach estimates parameter values for a virtual mixing console by analysing multitracks. These parameters are then applied to the multitracks to produce the final mix [18]. During training, a loss is calculated between the predicted mix and a ground-truth mix using audio or frequency domain metrics. However, this supervised training approach has inherent limitations, as mixing is not a one-to-one transformation [9]. For a given set of multitracks, there can be multiple “good” mixes, each influenced by the production context and the client’s intent [24]. This “intent” is typically communicated through various forms, including reference songs, demo tracks, semantic descriptors, or verbal instructions [24]. A reference song, in particular, is a professionally mixed track that serves as a stylistic guide, informing the desired tonal balance, spatialisation, and overall mix characteristics of the multitracks being processed. Recent research has explored context-aware mixing systems that leverage both DT and PE methods, bypassing the constraints of traditional supervised training by integrating user-provided contextual information. This approach has been named “mixing style transfer” in literature [8, 16, 22, 25].

The design of deep learning-based mixing systems must align with their intended use case and target users’ needs [23]. Recent literature identifies and categorizes users into three categories: amateurs, pro-amateurs (pro-ams), and professionals. Amateurs, with little to no mixing expertise, prefer fully automated systems requiring minimal control, while skilled users—pro-ams and professionals—seek tools offering extensive customisation and control. The primary difference between pro-ams and professionals lies in the commercial aspect, as professionals earn income from their mixing work. Amateurs are often satisfied with “good enough” mixes,

---

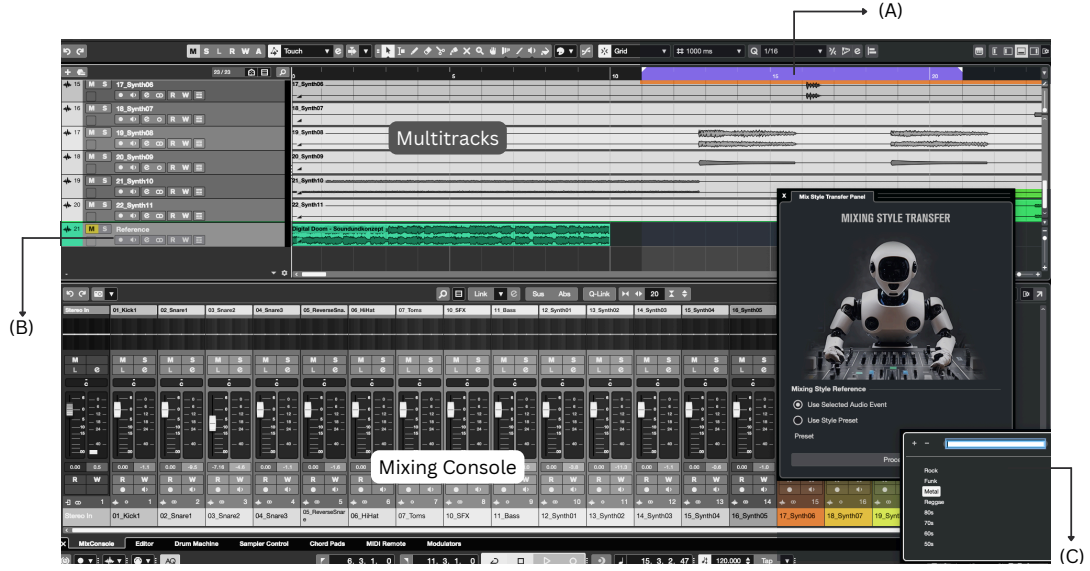
Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

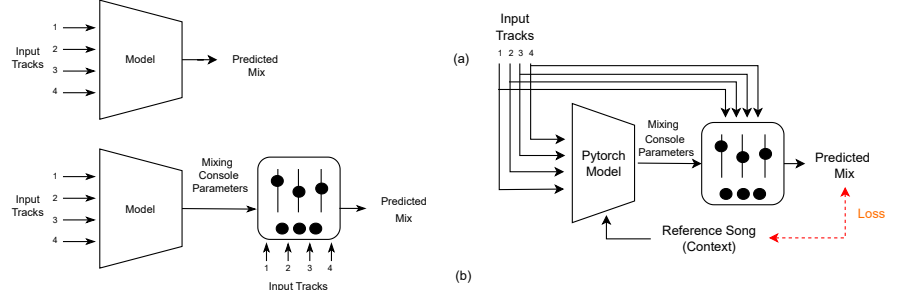
© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3721167>



**Figure 1: User interface for Diff-MSTC plugin integrated in DAW Cubase along with usage guidelines. (A) Select desired 10-sec section of input tracks for processing. (B) If you wish to use an audio event as reference, add it as one of the tracks and mute it. (C) The plugin with option to provide reference song using an audio event or select from already available preset associated with genres**



**Figure 2: (a) Direct estimation method (b) Parameter estimation method (c) Overview of Diff-MST training**

whereas skilled users demand high-quality results and seamless integration into existing workflows, highlighting the need for designing mixing systems that cater to varying levels of expertise and user requirements. RoEX<sup>1</sup>, a popular online platform for multitrack mixing favored by hobbyists, has added a DAW integration feature. However, users need to upload their tracks to the site for mixing prior to downloading a project file that is compatible with specific DAW software like Ableton, Bitwig, and Studio One. This approach still poses constraints on integration fluidity and overlooks essential mixing context.

In this demonstration, we present **Diff-MSTC**, a controllable and a reference song-based context-aware deep learning-based system for multitrack music mixing, seamlessly integrated into the DAW called Cubase<sup>2</sup>. We begin with an overview of the deep learning architecture and the integration pipeline within Cubase. We briefly

discuss the user-centric approach to the design and development of the deep learning system as well as the prototype. Next, we outline the steps for using the system in Cubase and share a first-person reflection on the interaction, emphasising experienced limitations. In Section 4, we explore broader implications of such a system on creative workflows. Finally, in Section 5, we propose future directions, focusing on a **human-centred iterative design approach**. A video walkthrough of the system can be found here<sup>3</sup>

### 3 Implementation

**Diff-MST** is a differentiable mixing style transfer model trained using a self-supervised training approach to transfer the style of a reference song to the provided multitrack. **Diff-MSTC** represents the integration of Diff-MST as a Steinberg Kernel Interface (SKI) plugin in Cubase. In this section, we provide an overview of Diff-MST and the integration methodology in Cubase.

<sup>1</sup><https://www.roexaudio.com/automix-pro>

<sup>2</sup><https://www.steinberg.net/cubase/>

<sup>3</sup>[https://www.youtube.com/watch?v=-wXFBgWi\\_tw](https://www.youtube.com/watch?v=-wXFBgWi_tw)

### 3.1 Diff-MST

We discuss the architecture, approach, training, and inference of our system, Diff-MST, as detailed in [25]. The system takes as input a multitrack and a reference song and predicts channel strip <sup>4</sup> parameters for each track as shown in Figure 3. The system is designed to be invariant to the number and order of tracks.

**3.1.1 Architecture.** The architecture of Diff-MST comprises a spectrogram encoder, a transformer controller, and a differentiable mixing console. **Spectrogram Encoder:** A shared-weight encoder is used to capture information from the tracks and the reference song. The encoder first converts the audio into the frequency domain using short-time Fourier transform (STFT) and then employs a CNN-based architecture to generate fixed-size embeddings. **Transformer Controller:** The embeddings are processed by a transformer encoder that performs style transfer through an attention mechanism. This is followed by a multilayer perceptron (MLP) that predicts parameters for audio effects in the mixing console. **Differentiable Mixing Console (DMC):** The DMC includes a channel strip on each track, which contains an audio effect chain comprising gain, a 4-band parametric equalizer, a dynamic range compressor, and panorama controls. The sum of the transformed tracks is sent to a master bus consisting of a dynamic range compressor, an equalizer, and an output fader. This master bus processing is essential since the provided reference songs are typically mastered. All effects in the console are implemented to be differentiable using the auto-differentiation method [4, 14, 16, 17]. The predicted parameters, along with the input tracks, are passed through the mixing console to produce the final mix in the target style. Using differentiable effects enables end-to-end training with backpropagation through the system.

**3.1.2 Training and Inference.** The system is trained using a self-supervised approach. We use multitrack datasets from MedleyDB [1, 2] and Cambridge.mt<sup>5</sup> and reference songs from the MTG Jambendo dataset [3]. A custom loss function computes the mean square error between a weighted average of audio features related to dynamics, spatialisation, and equalisation for the predicted mix and the reference song. During inference, the system processes a 10-second section of the input tracks and the reference song, generating predicted parameters and the resulting mix as output. The system produces a static mix with a fixed effect chain on each track.

### 3.2 Diff-MSTC

In previous work, we surveyed skilled mixing engineers and found strong interest in a controllable, context-aware mixing assistant [9] for creating demo mixes and serving as a starting point [23]. Diff-MST’s architecture was designed to meet these needs, with its integration shaped by Cubase workflows, infrastructure, and model constraints [26]. The Diff-MST model is integrated into Cubase as a **Steinberg Kernel Interface (SKI)** plugin. The SKI is a proprietary SDK that extends the functionality of Virtual Studio Technology (VST3). The PyTorch model is optimised for use within Cubase

by leveraging *TorchScript*<sup>6</sup>. Additionally, the user interface for the prototype is developed using **Steinberg’s VST3 SDK**<sup>7</sup>. The UI was developed using a user-centric design (UCD) approach. UCD follows an iterative approach that combines generative (brainstorming) and investigative (surveys and interviews) methods to understand users’ needs at each design phase [15]. We considered various use cases of the system across varied user-profile and skill levels. Thereafter, we determined their user requirements for the same and developed UI design for selecting the section of the tracks, providing reference songs, and preview of the results. The plugin interface includes a simple user interface (UI) featuring a robot mascot called “Mixxy” and options for specifying the reference song as shown in Figure 1.

## 4 Usage

To start, the user should launch the Cubase application and drag the multitrack files into the DAW. Next, they should load each audio file onto its own track. To utilize the Diff-MSTC prototype, the user must navigate to *Project > Mix Style Transfer* in the control panel on macOS. Then, a 10-second segment of the multitrack must be selected as input for the model using the playback cursor. It’s advised to pick a segment where all tracks are active, since the encoder’s embeddings rely on the chosen section. If any track’s segment is muted during selection, it will be excluded from processing. Similarly, any entirely muted track will not be processed. The interface offers two methods for selecting a reference song. The first method allows loading a personal reference song by adding its audio file to a new track and muting it. The user then selects the audio event and clicks *process*. For the second method, which uses preset reference songs, the user chooses a genre style from the presets and clicks *Process*. Figure 1 illustrates the steps for utilizing the system. The system will then evaluate each track and the reference song, applying the foreseen parameters to the mixing console. Figure 4 shows the state of the mixing console before and after processing. Further tweaks can be applied, if necessary, to refine the mix after playback.

### 4.1 First-Person Account: Reflections on Diff-MSTC’s Development and Use

Developing Diff-MSTC was as much about technical problem-solving as it was about understanding real-world workflows and user needs. As someone with experience in music mixing, I approached this project both as a researcher and a potential end-user, which shaped my perspective throughout the process.

**Usability Challenges:** One of the first challenges I encountered was the requirement to manually select a 10-second multitrack section where all tracks were active. While this step is necessary due to model constraints—such as the memory-intensive nature of training deep learning models with high-resolution multitrack audio—it felt cumbersome in practice. Automating this step or eliminating the need for manual selection altogether would significantly streamline the workflow.

Another key limitation arose during the integration of the model’s differentiable effects with the VST plugins in Cubase. Despite efforts to map parameters accurately, differences in how the effects

<sup>4</sup>A channel strip is a collection of audio processing units in a mixing console that allows users to process sound.

<sup>5</sup><https://cambridge-mt.com/>

<sup>6</sup><https://pytorch.org/docs/stable/jit.html>

<sup>7</sup><https://github.com/steinbergmedia/vst3sdk>

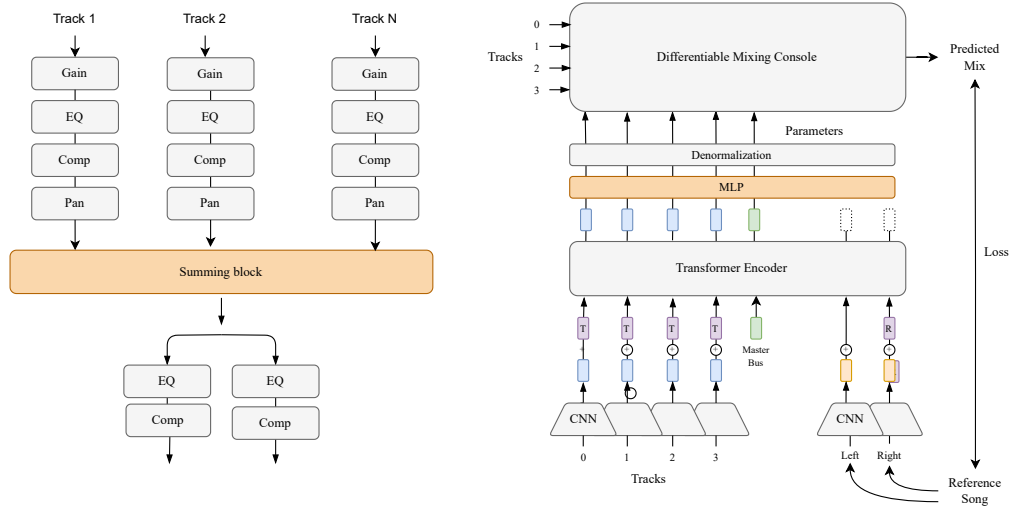


Figure 3: Differentiable mixing console (left) and Diff-MST architecture (right) [25]



Figure 4: Mixing console in Cubase before (left) and after (right) the style transfer

were implemented led to noticeable audio discrepancies. This made it difficult to achieve seamless transitions from the model’s outputs to Cubase’s native processing. Addressing this issue remains a challenge, whether through neural proxies [18], use of VST implementations of differentiable effects during inference, or gradient-free optimisation techniques [19]. The system applied impractical and extreme EQ curves on several instances due to the above mentioned technical limitation which hindered my experience with the usage.

Despite these hurdles, using Diff-MSTC to generate mix suggestions was a creatively rewarding experience. The ability to quickly explore mixes in different genres offered fresh perspectives on familiar material, often sparking new ideas. At the same time, I found myself wanting more control—such as the option to bypass specific effects or selectively apply processing to certain tracks. Finally, using the system within a real DAW environment provided insights that would have been impossible to gain in a purely theoretical setting.

## 5 Thoughts for Introspection

The integration of AI technologies into creative workflows raises critical questions about their interaction and impact. How effectively does Diff-MSTC assist users in their creative processes, and to what extent does it engage them? Are users satisfied enough with its output to use it with minimal adjustments, and how does this influence their sense of authorship? Conversely, if users find themselves frequently dissatisfied with the results, does the system still hold value in specific contexts? Under what circumstances does it serve as a catalyst for creativity rather than a hindrance? How do users adapt to and navigate its imperfections, and what emergent behaviours arise as they incorporate the system into their workflows? One of Diff-MSTC’s strengths lies in enabling rapid iteration, allowing users to explore multiple mix styles in a fraction of the traditional time. This capability fosters a more dynamic, exploratory workflow where users can experiment across genres and find unexpected creative directions. In this context, the system transitions from being merely a technical tool to a creative enabler. Additionally, Diff-MSTC has the potential to serve as an

educational tool, helping users analyse and learn from its decision-making processes. By exposing them to its parameter choices, the system may inspire new techniques and encourage innovation. It invites questions about how AI-driven tools can transform traditional practices, foster emergent workflows, and contribute to the evolution of creative fields like music production.

## 6 Conclusion and Future Directions

In this demonstration, we present **Diff-MSTC**, a prototype for a controllable and reference song-based context-aware multitrack mixing system integrated into the DAW Cubase. We provide an overview of the system and the integration process. Based on a first-person account, we discuss the challenges encountered during integration and emphasize the non-trivial process of transitioning from a deep learning model to an integrated prototype in our case. Finally, we highlight critical questions that this AI system raises regarding its role in creative tools and workflows.

As a research prototype, Diff-MSTC is inherently limited. We plan to conduct user experience and usability studies with participants of varying expertise to better understand its utility, value, and impact on workflows. This will enable us to build human-centered solutions, uncover further system limitations, and iteratively refine the prototype for improved efficiency and practical utility.

## Acknowledgments

This work is funded and supported by UK Research and Innovation [grant number EP/S022694/1] and Steinberg Media Technologies GmbH under the AI and Music Centre for Doctoral Training (AIM-CDT) at the Centre for Digital Music, Queen Mary University of London, London, UK. We would like to thank Franco Caspe for his feedback on the manuscript.

## References

- [1] Rachel M Bittner et al. 2016. MedleyDB 2.0: New data and a system for sustainable data collection. In *Proceedings of the 17th Int. Society for Music Information Retrieval Conference Late Breaking and Demo Papers*. Int. Society for Music Information Retrieval, New York, USA.
- [2] Rachel M Bittner, Justin Salamon, Mike Tierney, Matthias Mauch, Chris Cannam, and Juan Pablo Bello. 2014. MedleyDB: A multitrack dataset for annotation-intensive music research. In *Proc. of the 15th Int. Society for Music Information Retrieval Conf. (ISMIR)*, Vol. 14. Int. Society for Music Information Retrieval, Taipei, Taiwan, 155–160.
- [3] Dmitry Bogdanov, Minz Won, Philip Tovstogan, Alastair Porter, and Xavier Serra. 2019. The MTG-Jamendo Dataset for Automatic Music Tagging. In *Machine Learning for Music Discovery Workshop, International Conference on Machine Learning*. International Conference on Machine Learning, Long Beach, CA, United States.
- [4] Joseph T. Colonel, Christian J. Steinmetz, Marcus Michelen, and Joshua D. Reiss. 2022. Direct Design of Biquad Filter Cascades with Deep Learning by Sampling Random Polynomials. In *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Singapore, 3104–3108. doi:10.1109/ICASSP43922.2022.9747660
- [5] Brecht De Man and Joshua D Reiss. 2013. A knowledge-engineered autonomous mixing system. In *135th Audio Engineering Society Convention*. Audio Engineering Society, New York, USA.
- [6] Brecht De Man, Ryan Stables, and Joshua D Reiss. 2019. *Intelligent Music Production*. Taylor & Francis Group, UK.
- [7] Roey Izhaki. 2017. *Mixing Audio: Concepts, Practices, and Tools*. Routledge, UK.
- [8] Junghyun Koo et al. 2023. Music Mixing Style Transfer: A Contrastive Learning Approach to Disentangle Audio Effects. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Rhodes Island, Greece, 1–5.
- [9] M Nyssim Lefford, Gary Bromham, Gyorgy Fazekas, and David Moffat. 2021. Context-Aware Intelligent Mixing Systems. *Journal of the Audio Engineering Society* 69 (March 2021), 128–141.
- [10] Marco A Martínez-Ramírez, Wei-Hsiang Liao, Giorgio Fabbro, Stefan Uhlich, Chihiro Nagashima, and Yuki Mitsufuji. 2022. Automatic music mixing with deep learning and out-of-domain data. In *Proc. of the 23rd Int. Society for Music Information Retrieval Conf. (ISMIR)*. Int. Society for Music Information Retrieval, Bengaluru, India, 411–418.
- [11] Marco A Martínez-Ramírez, Daniel Stoller, and David Moffat. 2021. A Deep Learning Approach to Intelligent Drum Mixing With The Wave-U-Net. *Journal of the Audio Engineering Society* 69 (March 2021), 142–151.
- [12] Michael Miller. 2016. *Mixing Music*. Dorling Kindersley Ltd, London, UK.
- [13] Dave Moffat and Mark Sandler. 2019. Machine learning multitrack gain mixing of drums. In *147th Audio Engineering Society Convention*. Audio Engineering Society, Dublin, Ireland, UK.
- [14] Shahan Nercessian. 2020. Neural parametric equalizer matching using differentiable biquads. In *Proceedings of the International Conference on Digital Audio Effects 2020*. Proceedings of the International Conference on Digital Audio Effects, Vienna, Austria, 265–272.
- [15] Donald A. Norman (Ed.). 1986. *User Centered System Design: New Perspectives on Human-computer Interaction* (1st ed.). CRC Press, Hillsdale, USA. doi:10.1201/9780367807320
- [16] Christian J Steinmetz, Nicholas J Bryan, and Joshua D Reiss. 2022. Style Transfer of Audio Effects with Differentiable Signal Processing. *Journal of the Audio Engineering Society* 70, 9 (September 2022), 708–721.
- [17] Christian J Steinmetz, Vamsi Krishna Ithapu, and Paul Calamia. 2021. Filtered noise shaping for time domain room impulse response estimation from reverberant speech. In *2021 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics*. IEEE, Mohonk, USA, 221–225. doi:10.1109/WASPAA52581.2021.9632680
- [18] Christian J Steinmetz, Jordi Pons, Santiago Pascual, and Joan Serrà. 2021. Automatic multitrack mixing with a differentiable mixing console of neural audio effects. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Toronto, Ontario, Canada, 71–75. doi:10.1109/ICASSP39728.2021.9414364
- [19] Christian J. Steinmetz, Shubhr Singh, Marco Comunità, Ilias Ibnayyahya, Shanxin Yuan, Emmanouil Benetos, and Joshua D. Reiss. 2024. ST-ITO: Controlling Audio Effects for Style Transfer with Inference-Time Optimization. arXiv:2410.21233 [cs.SD] <https://arxiv.org/abs/2410.21233>
- [20] Christian J. Steinmetz, Soumya Sai Vanka, Marco A. Martínez Ramírez, and Gary Bromham. 2022. *Deep Learning for Automatic Mixing*. Proc. of the 23rd Int. Society for Music Information Retrieval Conf. (ISMIR), Bengaluru, India. <https://dl4am.github.io/tutorial>
- [21] Ajin Tom, Joshua D Reiss, and Philippe Depalle. 2019. An automatic mixing system for multitrack spatialization for stereo based on unmasking and best panning practices. In *146th Audio Engineering Society Convention*. Audio Engineering Society, Dublin, Ireland, UK.
- [22] Soumya Vanka, Jean-Baptiste Rolland, and György Fazekas. 2021. Intelligent Music Production: Music Production Style Transfer and Analysis of Mix Similarity. In *16th Digital Music Research Network (DMRN+ 16) Workshop*. Centre for Digital Music, QMUL, London, UK.
- [23] Soumya Vanka, Maryam Safi, Jean-Baptiste Rolland, and György Fazekas. 2023. Adoption of AI Technology in Music Mixing Workflow: An Investigation. In *154th Audio Engineering Society Convention*. Audio Engineering Society, Espoo, Finland.
- [24] Soumya Vanka, Maryam Safi, Jean-Baptiste Rolland, and György Fazekas. 2023. The Role of Communication and Reference Songs in the Mixing Process: Insights from Professional Mix Engineers. arXiv:2309.03404 [cs.HC]
- [25] Soumya Vanka, Christian Steinmetz, Jean-Baptiste Rolland, Joshua Reiss, and György Fazekas. 2024. Diff-MST: Differentiable Mixing Style Transfer. In *Proc. of the 25th Int. Society for Music Information Retrieval Conf. (ISMIR)*. Int. Society for Music Information Retrieval (ISMIR), San Francisco, USA, 563–570.
- [26] Soumya Sai Vanka, Lennart Hannink, Jean-Baptiste Rolland, and George Fazekas. 2024. Diff-MSTC: A Mixing Style Transfer Prototype for Cubase. arXiv:2411.06576 [eess.AS] <https://arxiv.org/abs/2411.06576>